

Self Learning Assignment: regional_sales_data

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```
In [14]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sys
```

```
In [15]: df = pd.read_csv('regional_sales_data.txt', sep=' ')
df.head()
```

```
Out[15]:
```

	Region_ID	Units_Sold	Unit_Price	Store_name	Country
0	REG_01	NaN	34.75	Beta_Retail	UK
1	REG_01	149.0	17.82	Delta_Shop	India
2	REG_01	NaN	30.94	Epsilon_Hub	India
3	REG_02	149.0	47.75	Delta_Shop	Canada
4	REG_03	11.0	78.29	Beta_Retail	Germany

1. Check and replace all missing values in the Units_Sold column with the median value of that specific column

```
In [16]: med = np.median(df['Units_Sold'].dropna())
print(f'median is {med}')
df['Units_Sold'] = df['Units_Sold'].fillna(med)

# firstly find median of units sold by removing null values
#
#
# Then filled null values with median value
```

median is 100.0

```
In [ ]:
```

2. Calculate the total units sold and the average unit price for each Store_name. Identify which store has generated the highest total units sold

```
In [17]: df_agg = df.groupby('Store_name').agg({'Unit_Price':['mean'], 'Units_Sold':['sum']})
print(df_agg)
```

```
df_agg.columns = ['Store_name', 'Price Average', 'Total Units Sold']
df_agg

# firstly grouped dataset with Store_name
# then applied mean function on Unit_Price column, and sum on Units Sold
```

	Store_name	Unit_Price mean	Units_Sold sum
0	Alpha_Mart	62.286667	881.0
1	Beta_Retail	59.816667	1042.0
2	Delta_Shop	40.494000	1001.0
3	Epsilon_Hub	59.321333	1376.0
4	Gamma_Store	55.271818	1057.0

Out[17]:

	Store_name	Price Average	Total Units Sold
0	Alpha_Mart	62.286667	881.0
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In []:

In [18]:

```
df_agg[df_agg['Total Units Sold']== max(df_agg['Total Units Sold'])]

# done filtering of Total Units Sold by maximum value Total Units Sold
```

Out[18]:

	Store_name	Price Average	Total Units Sold
3	Epsilon_Hub	59.321333	1376.0

In [19]:

```
df.head(2)
```

Out[19]:

	Region_ID	Units_Sold	Unit_Price	Store_name	Country
0	REG_01	100.0	34.75	Beta_Retail	UK
1	REG_01	149.0	17.82	Delta_Shop	India

3. Create a new column named Total_Revenue, calculated by multiplying Units_Sold by Unit_Price

In [20]:

```
df['Total_Revenue'] = df['Units_Sold'] * df['Unit_Price']

...
extracted Units_Sold and Unit_Price series
then cal multiply them
```

```
stored in total_revenue series column of df
'''
```

```
Out[20]: ' \nexttracted Units_Sold and Unit_Price series\nthen cal multiply them\nstored i
n total_revenue series column of df\n'
```

```
In [21]: df.head()
```

```
Out[21]:
```

	Region_ID	Units_Sold	Unit_Price	Store_name	Country	Total_Revenue
0	REG_01	100.0	34.75	Beta_Retail	UK	3475.00
1	REG_01	149.0	17.82	Delta_Shop	India	2655.18
2	REG_01	100.0	30.94	Epsilon_Hub	India	3094.00
3	REG_02	149.0	47.75	Delta_Shop	Canada	7114.75
4	REG_03	11.0	78.29	Beta_Retail	Germany	861.19

4. Group the data by Region_ID and calculate the sum of Total_Revenue for each region

```
In [22]: '''
drop null values of total revenue
then group by region id
calculate sum of total revenue for each group
reset index of group and add columns so they can be read easily
'''

df_agg = df.groupby('Region_ID').agg({'Total_Revenue':['sum']}).reset_index()

df_agg.columns = ['Region_ID', 'Total_Revenue']
df_agg
```

```
Out[22]:
```

	Region_ID	Total_Revenue
0	REG_01	117092.28
1	REG_02	63404.82
2	REG_03	62422.10
3	REG_04	52051.47

```
In [ ]:
```

5. Use Matplotlib to plot a bar chart showing Region_ID on the x-axis and the sum of Total_Revenue on the y-axis. Set the bar colour to green, add a grid on the y-axis, and give the chart a clear title

```
In [23]: df.head()
```

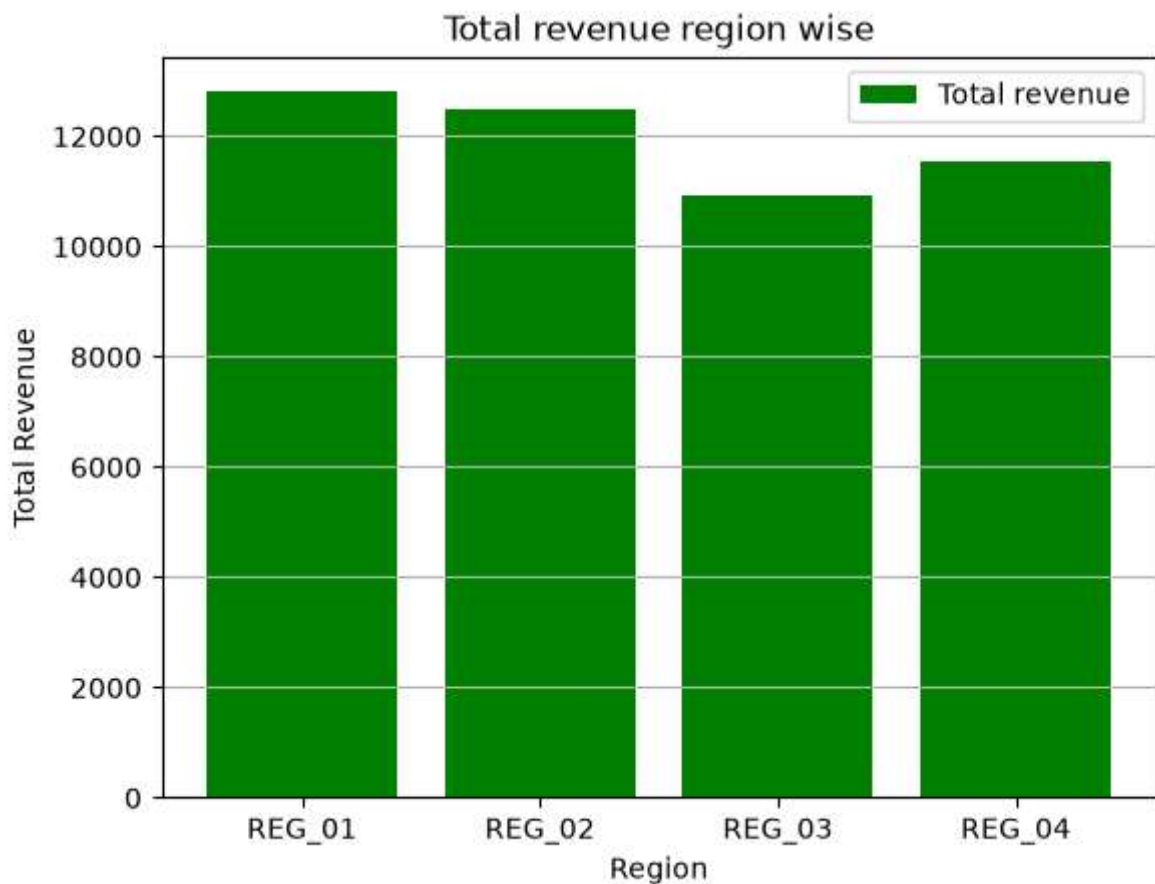
Out[23]:

	Region_ID	Units_Sold	Unit_Price	Store_name	Country	Total_Revenue
0	REG_01	100.0	34.75	Beta_Retail	UK	3475.00
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In [24]:

```
plt.bar(df['Region_ID'],df['Total_Revenue'] , label='Total revenue',color='green' )
plt.grid(axis='y')
plt.legend()
plt.xlabel('Region')
plt.ylabel('Total Revenue')
plt.title('Total revenue region wise')
plt.show()

'''
This graph shows region on x axis and total revenue on y axis
then set title, added with xlabel and ylabel
'''
```



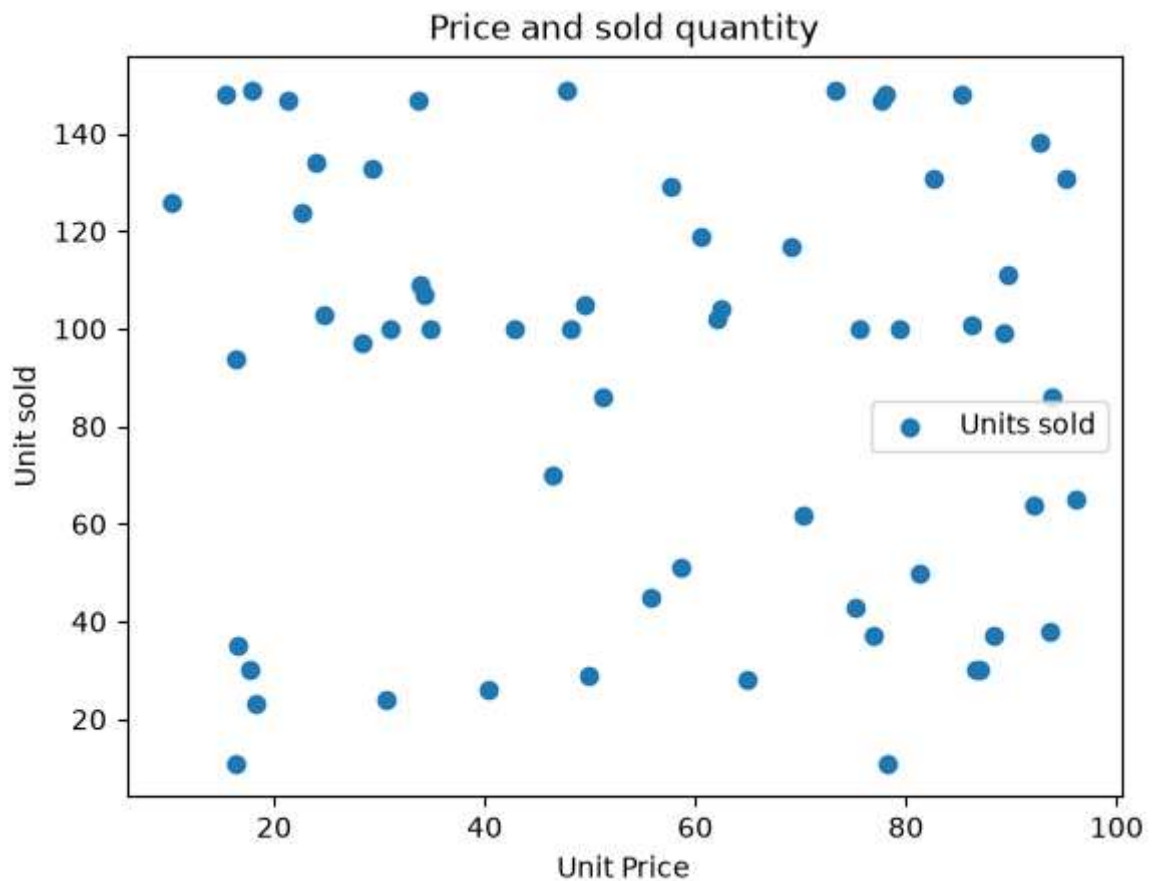
Out[24]: ' \nThis graph shows region on x axis and total revenue on y axis\nthen set title, added with xlabel and ylabel\n'

6. Create a scatter plot using Matplotlib mapping Unit_Price on the x-axis against Units_Sold on the y-axis

```
In [25]: plt.scatter(x = df['Unit_Price'],y= df['Units_Sold'], label='Units sold')
plt.xlabel('Unit Price')
plt.title('Price and sold quantity')
plt.legend()
plt.ylabel('Unit sold')

'''
This graph shows unit price on x axis and unit sold on y axis
then set title, added with xlabel and ylabel
'''
```

```
Out[25]: ' \nThis graph shows unit price on x axis and unit sold on y axis\nthen set titl
e, added with xlabel and ylabel\n'
```



```
In [ ]:
```